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Keeping GenAI Data Reliable: Vector Maintenance and Amazon S3 Storage & Security

Updating Vector Stores, Reranking Results, Managing S3
Storage Classes, Lifecycle, Replication, Encryption,
Logs, and Access Points

Why This Topic Matters



In the previous videos, I studied different places where a Generative AI application can store and search information, including OpenSearch, S3 Vectors, Aurora, and DynamoDB.



But **storing the data one time is not enough**. Real business information keeps changing. New documents are uploaded, old information is replaced, product details change, and new conversations are created.



If the vector store is not updated, the AI system may continue retrieving old information even when the original source has already changed.



At the same time, companies also need to think about **storage cost, security, backup, encryption, and who is allowed to access the data**.



So this video focuses on the operational side of GenAI data: keeping vector information fresh and managing the underlying S3 data correctly.

Keeping a Vector Store Up to Date



A VECTOR STORE NEEDS TO STAY SYNCHRONIZED WITH THE ORIGINAL DATA SOURCE. IF A COMPANY CHANGES A POLICY DOCUMENT, THE VECTOR STORE SHOULD EVENTUALLY REFLECT THAT NEW VERSION INSTEAD OF CONTINUING TO RETRIEVE EMBEDDINGS CREATED FROM THE OLD DOCUMENT.



THE COURSE INTRODUCES SEVERAL WAYS TO HANDLE THIS, INCLUDING **INCREMENTAL UPDATES, REAL-TIME CHANGE DETECTION, AUTOMATED SYNCHRONIZATION, AND SCHEDULED REFRESH PIPELINES.**



INCREMENTAL UPDATES MEAN WE UPDATE ONLY THE INFORMATION THAT CHANGED INSTEAD OF REBUILDING EVERYTHING EVERY TIME. THIS CAN SAVE PROCESSING TIME WHEN MOST OF THE DATA STAYS THE SAME.



REAL-TIME CHANGE DETECTION IS USEFUL WHEN NEW OR CHANGED INFORMATION NEEDS TO BECOME SEARCHABLE QUICKLY. SCHEDULED REFRESHES ARE MORE SUITABLE WHEN THE APPLICATION CAN WAIT AND UPDATE AT A FIXED TIME.



AMAZON EVENTBRIDGE CAN HELP AUTOMATE THESE UPDATE WORKFLOWS BY TRIGGERING ACTIONS WHEN AN EVENT OCCURS OR ACCORDING TO A SCHEDULE.

Example: Updating a GenAI Knowledge Source

- Imagine a company stores HR documents in Amazon S3.
- **What happens when a document changes?**
 1. A new version of the employee policy is uploaded.
 2. The system detects the change.
 3. A workflow is triggered using services such as EventBridge and Lambda.
 4. The changed document is processed again.
 5. New embeddings are created.
 6. The vector store is updated.
- **Why is this important?**
 - If the source document says employees now receive **18 vacation days**, but the vector store still contains the old **15-day policy**, the AI assistant could give the wrong answer.
 - The goal is to keep:
 - **Source Data → Embeddings → Vector Store**
 - as closely synchronized as possible.



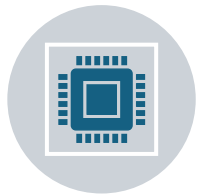
Vector Index Maintenance Over Time

- Even if data is updated correctly, a vector index may still need maintenance.
- **The course explains that drift and fragmentation can become problems over time.** As data is continuously added, changed, or removed, the index may no longer be organized as efficiently as it was when first created.
- One approach is to periodically rebuild the vector index instead of keeping the same index forever.
- EventBridge can trigger a scheduled maintenance process, while **AWS Batch can run larger background jobs** needed to rebuild the vector data.
- **Example Maintenance Flow**
- **EventBridge Trigger**
 - **AWS Batch Job**
 - Create New Embeddings
 - Create New Vector Database
 - Validate New Index
 - Replace Old Index with New One
- The important part is **validation before switching**. A new index should be checked before replacing the working version so the application does not move to a broken or incomplete dataset.

Why Reranking Can Improve Retrieval

- Vector search may return several relevant chunks, but the first result is not always the most useful one.
- **A reranker looks at the retrieved results again and decides which chunks are most relevant to the actual query.**
- For example, a vector search may return ten sections that are all related to employee benefits. A reranker can place the section that most directly answers the user's question at the top.
- Amazon Bedrock supports reranking through a **Rerank operation**, and the course also explains that a reranker model can be specified when using a Knowledge Base.
- The reranker does not replace the vector store. Instead, it works **after retrieval** to improve the order of the returned information.
- **Simple Flow**
- **Query**
 - Vector Search
 - Several Relevant Chunks
 - Reranker
 - Best Chunks First
 - Foundation Model
- This can improve RAG quality because the model receives the most useful context earlier in the retrieved results.

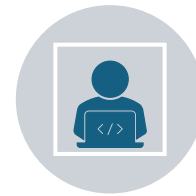
Why Amazon S3 Is Important in GenAI



Amazon S3 is not a vector database in the traditional sense, but it is still a very important part of many Generative AI systems.



S3 is commonly used to store the original source data such as PDFs, text files, reports, images, audio files, logs, and training datasets.



A Knowledge Base or data-processing pipeline may read files from S3, prepare them, create embeddings, and then send those embeddings into a vector store.



Because companies may keep very large amounts of AI-related data, **storage cost becomes important over time.**



Some files may be used every day, while others may be needed only a few times per year.



This is why S3 provides several storage classes instead of treating every object the same way.

S3 Storage Classes: Different Data, Different Cost

- The course covers several major S3 storage classes:
- **S3 Standard**
- **S3 Standard-Infrequent Access**
- **S3 One Zone-Infrequent Access**
- **S3 Glacier Instant Retrieval**
- **S3 Glacier Flexible Retrieval**
- **S3 Glacier Deep Archive**
- **S3 Intelligent-Tiering**
- These classes are designed for **different access patterns and cost requirements**.
- A file that is read every day should not necessarily be stored in the same class as a compliance archive that may not be opened for several years.
- Objects can also move between storage classes later, either manually or automatically through **Lifecycle configurations**.
- **Main idea:**
- The correct storage class depends mainly on **how often the data is needed and how quickly it must be retrieved**.

Durability vs. Availability

- These two terms sound similar, but they describe different risks.
- **Durability**
- **Durability is about whether the object is preserved over time.**
- The course states that S3 provides **99.999999999% durability, or eleven 9s**, across its storage classes.
- This means S3 is designed so the chance of permanently losing an object is extremely small.
- **Availability**
- **Availability is about whether the object can be accessed when we request it.**
- Availability can be different depending on the storage class.
- **Easy Way to Remember**
- **Durability:** “Will my file still exist?”
- **Availability:** “Can I access my file right now?”
- This distinction is important because a storage class can have very high durability while offering a different availability level.

S3 Standard and Infrequent Access

S3 Standard

S3 Standard is designed for frequently accessed data.

It provides low latency and high throughput, which makes it suitable when files need to be accessed often.

In a GenAI application, this could include active documents used regularly by a Knowledge Base or current datasets used by an application.

S3 Standard-IA

Standard-Infrequent Access is designed for data that is accessed less often but still needs fast access when requested.

The storage cost is lower than Standard, but the access pattern should really be infrequent.

S3 One Zone-IA

One Zone-IA keeps data in a **single Availability Zone** instead of multiple AZs.

This can make sense for secondary copies or information that can be recreated if necessary.

Example

Current product manuals: S3 Standard

Older but still needed manuals: Standard-IA

Re-creatable processed files: One Zone-IA

Amazon S3 Glacier Storage Classes

- Glacier storage classes are mainly designed for **archiving and backup**.
- **Glacier Instant Retrieval**
 - Used when archived data is not accessed often, but the application still needs **millisecond retrieval** when it is requested.
- **Glacier Flexible Retrieval**
 - Lower-cost archive storage where retrieval can take longer.
 - The course shows different retrieval speeds such as expedited, standard, and bulk retrieval.
- **Glacier Deep Archive**
 - Designed for **very long-term storage** where data is rarely needed.
 - It provides very low storage cost, but retrieval can take many hours.
- **GenAI Example**
 - A company may keep:
 - Active training documents in S3 Standard
 - Older reference material in Standard-IA
 - Historical AI training datasets or compliance archives in Glacier
 - The data still exists, but the company does not pay the same storage price for information that is rarely accessed.

S3 Intelligent-Tiering

- Sometimes a company does not know how often a file will be used.
- **S3 Intelligent-Tiering monitors access patterns and automatically moves objects between access tiers.**
- Frequently used objects remain in the Frequent Access tier.
- Objects that are not accessed for a period of time can automatically move into lower-cost tiers.
- The course also includes optional archive and deep archive tiers for data that remains unused for much longer periods.
- Intelligent-Tiering has a monitoring and auto-tiering fee, but it can be useful when the access pattern is unpredictable.
- **Example**
- A company uploads thousands of technical documents.
- Some may become popular because of a new product release, while others may not be opened for months.
- Instead of guessing the correct storage class for each document, Intelligent-Tiering can adjust the storage tier based on actual usage.

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Automating Storage with S3 Lifecycle Rules

- S3 Lifecycle Rules allow storage management to happen automatically.
- **Transition actions move objects into another storage class after a defined amount of time.**
- For example, a file could start in S3 Standard and move to Standard-IA after 60 days.
- Later, the same object could move into Glacier for long-term archiving.
- **Expiration actions automatically delete objects when they are no longer required.**
- Lifecycle Rules can also clean up old file versions or incomplete multipart uploads.
- Rules can target only certain objects using prefixes or tags.
- **Example GenAI Policy**
- **0–30 days:** S3 Standard
- **31–180 days:** Standard-IA
- **After 180 days:** Glacier
- **After 5 years:** Delete if company policy allows it
- This can control cost without requiring someone to manually move thousands of files every month.

S3 Replication: CRR and SRR

- Amazon S3 can automatically replicate objects into another bucket.
- **Cross-Region Replication – CRR**
- Copies objects into a bucket in a **different AWS Region**.
- Possible uses include compliance, lower-latency access in another geographic area, or maintaining copies across accounts.
- **Same-Region Replication – SRR**
- Copies objects into another bucket in the **same AWS Region**.
- The course gives examples such as log aggregation or copying production data into another environment.
- **Important Requirements**
- Versioning must be enabled on both the source and destination buckets.
- Replication happens asynchronously.
- S3 needs the correct IAM permissions to perform the copy.
- In a GenAI system, replication can help protect important source documents or make copies available for separate environments.

S3 Encryption: Four Main Approaches

The course explains four main ways to protect S3 objects with encryption.

SSE-S3

Amazon S3 manages the encryption keys.

This is the simplest server-side encryption approach and is enabled by default for new S3 objects.

SSE-KMS

Uses keys managed through **AWS Key Management Service**.

This provides more control over keys and allows key usage to be audited.

SSE-C

The customer provides the encryption key.

Amazon S3 performs the encryption, but S3 does not store the customer-provided key.

Client-Side Encryption

The data is encrypted **before it is sent to S3**.

The customer controls both the encryption process and the keys.

The easiest way to separate these methods is to ask:

Who owns and manages the encryption key?

SSE-KMS and KMS Request Limits

- SSE-KMS gives more key control, but it also adds another AWS service into the request path.
- When an object is uploaded using SSE-KMS, S3 may call the **GenerateDataKey** KMS API.
- When an encrypted object is downloaded, the **Decrypt** API may be called.
- These requests count against AWS KMS service quotas.
- If an application has very high S3 request volume, KMS limits may become part of the performance design.
- The course notes that quota increases can be requested through the Service Quotas console.
- This is a good example of why security choices can also affect **performance and architecture**, not only encryption.

Encryption in Transit



Encryption at rest protects stored files, but the data should also be protected while it moves across the network.



Encryption in transit uses SSL/TLS, which is normally provided through HTTPS.



S3 supports both HTTP and HTTPS endpoints, but HTTPS provides encryption while the data is moving between the client and S3.



The course recommends HTTPS and notes that it is required when using SSE-C.



Easy Difference



Encryption at rest: protects the file while it is stored.



Encryption in transit: protects the file while it is traveling between systems.



A secure GenAI data pipeline usually needs both kinds of protection.

S3 Access Logs

- Security is not only about preventing access. Organizations also need to know **who tried to access the data**.
 - **S3 Access Logging can record requests made to a bucket**, including requests that were authorized and requests that were denied.
 - The logs are written into another S3 bucket where they can later be analyzed.
 - This can support auditing, troubleshooting, and security investigations.
 - The course notes that the target logging bucket must be in the same AWS Region.
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- **Important Warning**
 - The logging bucket should **not be the same bucket being monitored**.
 - If a bucket writes access logs back into itself, every new log creates another access event, which can create a logging loop and continuously increase the number of objects.
 - So a separate logging bucket should be used.

S3 Access Points and a GenAI Example

- When many teams or applications use the same S3 bucket, one large bucket policy can become difficult to manage.
- **S3 Access Points provide separate access paths to the same bucket.**
- Each Access Point has its own DNS name and its own access-point policy.
- For example, the Finance team could have an Access Point that allows access only to /finance/, while the Sales team has a different Access Point for /sales/.
- An Access Point can also be configured with **VPC Origin**, which means access is limited to resources inside a VPC.
- In that case, a VPC endpoint is needed, and the endpoint policy must allow access to the target bucket and Access Point.
- **Complete GenAI Example**
- Imagine a company builds an internal HR assistant.
- **S3** stores HR documents.
- **Lifecycle Rules** move older documents into cheaper storage.
- **Encryption** protects sensitive employee files.
- **Access Points** allow only the HR application to read the correct folder.
- **EventBridge** detects updates.
- **New embeddings** are created.
- **Vector store** is updated.
- **Reranking** improves the order of retrieved chunks.
- **Bedrock** generates the final answer.
- This shows how storage, security, automation, and retrieval all connect in one real GenAI architecture.

Key Takeaways & Thank You

- **Key Takeaways**
- **Vector stores need maintenance after deployment.** New or changed information must be synchronized, and large indexes may eventually need rebuilding.
- **Reranking can improve RAG retrieval** by reorganizing retrieved chunks according to how relevant they are to the user's question.
- Amazon S3 provides different storage classes so active and archived GenAI data do not need to be stored at the same cost level.
- **Lifecycle Rules automate movement and deletion**, while replication can create additional copies in the same or another AWS Region.
- Encryption can protect S3 data both **at rest and in transit**, and different encryption methods provide different levels of key control.
- Access logs improve auditing, while **S3 Access Points simplify access management** when different applications or teams use the same bucket.
- **Main Lesson**
- A reliable Generative AI application needs more than a model and a vector database.
- The data behind the model must remain:
- **Current → Relevant → Cost-Efficient → Protected → Accessible to the Right Users**
- **Thank You**
- **Prepared by: Akash Patel**
- **Concordia University Chicago**
- **Special thanks to Professor Dr. Victor Govindaswamy for his continued guidance and support throughout this course. This part of the course helped me understand that building a real Generative AI application is not only about generating responses. The data also needs to be updated, stored at the right cost, protected with proper security, and managed throughout its lifecycle. I appreciate his guidance in helping us connect these AWS services into complete and practical cloud architectures.**